

SQL Analytical Tasks

Step-by-Step Guide

Tank54 Group | Oracle HR Schema

What is in this book?

This guide has two parts. **Part 1** shows 5 examples. Each example has a task, step-by-step explanation, and the SQL query. **Part 2** has 15 practice tasks for you to solve.

Topics: SELECT, WHERE, ORDER BY, Aggregate Functions, GROUP BY, HAVING, CASE WHEN

Oracle HR Schema - EMPLOYEES Table

We use the Oracle HR database. For all tasks in this guide, we use only the **EMPLOYEES** table. This table has information about every employee in the company.

Column	Type	Example	Description
EMPLOYEE_ID	Number	100	Unique ID for each employee
FIRST_NAME	Text	Steven	First name of the employee
LAST_NAME	Text	King	Last name of the employee
EMAIL	Text	SKING	Email of the employee
PHONE_NUMBER	Text	515.123.4567	Phone number
HIRE_DATE	Date	17 - JUN - 03	Date when the employee started
JOB_ID	Text	AD_PRES	Job role code
SALARY	Number	24000	Monthly salary
COMMISSION_PCT	Number	0.3	Commission percent (some jobs)
MANAGER_ID	Number	(null)	ID of the employee's manager
DEPARTMENT_ID	Number	90	ID of the department

Quick Reminder - Key Concepts

Concept	What it does	Example
COUNT (*)	Counts rows	SELECT COUNT(*) FROM employees
SUM(column)	Adds all values	SELECT SUM(salary) FROM employees
AVG(column)	Calculates average	SELECT AVG(salary) FROM employees
MIN(column)	Finds smallest value	SELECT MIN(salary) FROM employees
MAX(column)	Finds largest value	SELECT MAX(salary) FROM employees
GROUP BY	Groups rows by a column	GROUP BY department_id
HAVING	Filters after GROUP BY	HAVING COUNT(*) > 5
WHERE	Filters rows before grouping	WHERE salary > 10000
ORDER BY	Sorts the result	ORDER BY salary DESC
CASE WHEN	Conditional logic	CASE WHEN salary > 10000 THEN 'High' ELSE 'Low' END

PART 1: Examples

In this part, we show 5 examples. Each example has 3 parts: the task, the steps, and the SQL query. Read each step carefully. Try to understand the logic before you look at the SQL code. All examples use only the EMPLOYEES table. No JOINS are used.

Example 1: Employee Count by Department

Task:

Count how many employees work in each department. Show department ID and the count. Show only departments with more than 5 employees.

Step-by-Step:

Step 1: What do we want to count? Employees in each department. So we need COUNT(*) to count rows.

Step 2: We need to count by department, so we use GROUP BY department_id. This puts employees into groups by their department.

Step 3: We only want departments with more than 5 employees. Because 5 is about the COUNT result, we use HAVING (not WHERE).

Step 4: Remember: WHERE filters rows before grouping. HAVING filters groups after grouping.

SQL Query:

```
SELECT department_id,
       COUNT(*) AS employee_count
FROM employees
GROUP BY department_id
HAVING COUNT(*) > 5
ORDER BY employee_count DESC;
```

Example 2: Average, Min and Max Salary

Task:

Find the average, minimum, and maximum salary for each job. Show job ID and all three values. Order by average salary from high to low.

Step-by-Step:

Step 1: We need three aggregate functions: AVG(salary), MIN(salary), and MAX(salary). All three work on the SALARY column.

Step 2: We want these values for each job, so we use GROUP BY job_id.

Step 3: We use ROUND() to make the average salary show only 2 decimal places. This makes the result cleaner.

Step 4: We order by the average salary in DESC order (high to low). We can use the alias "avg_sal" in ORDER BY.

SQL Query:

```
SELECT job_id,
       ROUND(AVG(salary), 2) AS avg_sal,
       MIN(salary) AS min_sal,
       MAX(salary) AS max_sal
FROM employees
GROUP BY job_id
ORDER BY avg_sal DESC;
```

Example 3: Employees with Salary in a Range

Task:

Find all employees whose salary is between 5000 and 15000. Show their full name, job ID, and salary. Order by salary from high to low.

Step-by-Step:

Step 1: We need a range filter: salary must be between 5000 and 15000. We can use BETWEEN for this.

Step 2: BETWEEN includes both numbers. So salary >= 5000 AND salary <= 1500. This is the same as BETWEEN 5000 AND 15000.

Step 3: We need the full name. We use || to join first_name and last_name with a space in between.

Step 4: We also want the job_id column to see what job each employee has.

SQL Query:

```
SELECT first_name || ' ' || last_name AS employee_name,
       job_id,
       salary
FROM employees
WHERE salary BETWEEN 5000 AND 15000
ORDER BY salary DESC;
```

Example 4: Salary Ranges - Counting Groups

Task:

Put employees into 3 salary groups: Low (salary < 5000), Medium (5000 to 10000), High (> 10000). Count how many employees are in each group.

Step-by-Step:

Step 1: We need to create a group label for each employee based on their salary. We use CASE WHEN for this.

Step 2: CASE WHEN salary < 5000 THEN 'Low' WHEN salary <= 10000 THEN 'Medium' ELSE 'High' END. The conditions check from top to bottom.

Step 3: To group by the CASE result, we write the full CASE expression in GROUP BY (not the alias). In Oracle, we can use the alias in ORDER BY.

Step 4: COUNT(*) counts the rows in each group. We order the result by count from high to low.

SQL Query:

```
SELECT CASE
    WHEN salary < 5000 THEN 'Low'
    WHEN salary <= 10000 THEN 'Medium'
    ELSE 'High'
END AS salary_group,
    COUNT(*) AS employee_count
FROM employees
GROUP BY CASE
    WHEN salary < 5000 THEN 'Low'
    WHEN salary <= 10000 THEN 'Medium'
    ELSE 'High'
END
ORDER BY employee_count DESC;
```

Example 5: Total Salary Cost per Department

Task:

Find the total salary cost for each department. Also find how many employees work in each department. Show department ID, employee count, and total salary. Order by total salary from high to low.

Step-by-Step:

Step 1: We need two things: the count of employees and the sum of salaries. We use COUNT(*) and SUM(salary).

Step 2: We group by department_id because we want these numbers for each department.

Step 3: We exclude departments with no employees by checking department_id IS NOT NULL in the WHERE clause.

Step 4: We order by the total salary (SUM result) from high to low to see which department costs the most.

SQL Query:

```
SELECT department_id,
    COUNT(*) AS employee_count,
    SUM(salary) AS total_salary
FROM employees
WHERE department_id IS NOT NULL
GROUP BY department_id
ORDER BY total_salary DESC;
```

PART 2: Practice Tasks

Now it is your turn! Solve these 15 tasks. Write your SQL queries on paper or in a database tool. Use only the EMPLOYEES table. No JOINs are needed. Good luck!

Task 1	Total Salary Cost of the Company	SUM
Find the total salary cost of the whole company. This means: add up all employee salaries. Show the total as one number.		
Task 2	How Many Employees in the Company?	COUNT
Count the total number of employees in the company. Show the count as one number.		
Task 3	Salary Statistics	AVG + MIN + MAX
Find the average, minimum, and maximum salary in the whole company. Show all three values in one result.		
Task 4	Employee Count by Department	GROUP BY + COUNT
Count how many employees are in each department. Show department ID and the count. Order by count from high to low.		
Task 5	Average Salary by Job	GROUP BY + AVG + ORDER BY
Find the average salary for each job ID. Show job ID and average salary (round to 2 decimals). Order by average salary from high to low.		
Task 6	Departments with More Than 10 Employees	GROUP BY + HAVING + COUNT
Find departments that have more than 10 employees. Show department ID and the count. Order by count from high to low.		
Task 7	High-Earning Jobs	GROUP BY + HAVING + AVG
Find jobs where the average salary is more than 10000. Show job ID and average salary. Order by average salary from high to low.		
Task 8	Employees with High Salary	WHERE + ORDER BY
Find all employees whose salary is more than 15000. Show their full name, job ID, and salary. Order by salary from high to low.		
Task 9	Commission vs No-Commission Employees	CASE WHEN + COUNT
Count how many employees have a commission and how many do not. Use CASE WHEN to create two groups: Has Commission and No Commission. Show both counts.		
Task 10	Employees Hired in a Specific Year	WHERE + Date Filter + ORDER BY
Find all employees who were hired in the year 2005. Show their full name, hire date, and salary. Order by hire date from old to new.		
Task 11	Commission Employees	WHERE IS NOT NULL + COUNT
Find all employees who have a commission (commission_pct is not null). Count how many such employees exist. Show the count.		
Task 12	Salary Ranges - Low, Medium, High	CASE WHEN + GROUP BY + COUNT
Group employees into 3 salary ranges: Low (less than 5000), Medium (5000 to 10000), High (more than 10000). Count how many employees are in each range. Use CASE WHEN.		
Task 13	Employees per Manager	GROUP BY + HAVING + COUNT
Count how many employees each manager has. Show manager ID and the count. Show only managers who have more than 5 employees. Order by count from high to low.		
Task 14	Highest Salary per Department	GROUP BY + MAX + ORDER BY
Find the highest salary in each department. Show department ID and the maximum salary. Order by maximum salary from high to low.		

Task 15**Departments with Highest and Lowest Salary Gap***GROUP BY + MAX - MIN + HAVING*

For each department, find the difference between the highest and lowest salary. Show department ID and the salary gap. Show only departments where the gap is more than 8000. Order by gap from big to small.